MATHEMATICS

Grade 7

26 March 2024

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SECTION A

*Answer **all** questions **[25 marks]***

1. **(a)** Write nineteen in Arabic figures.

(b) Express 1 030 in expanded index notation.

a._____ **(1)**

b._____

_____ **(1)**

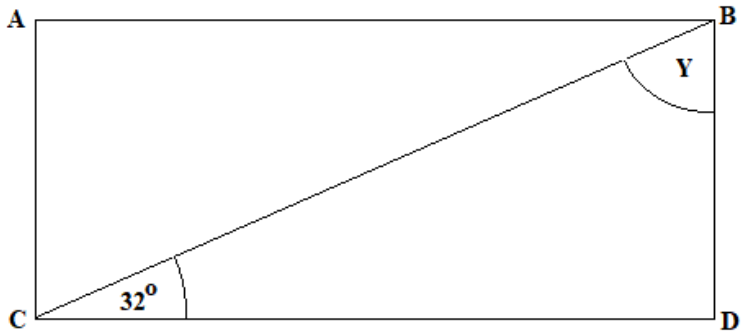
2 **(a)** Express 0,125 as a proper fraction in its lowest terms.

(b) Simplify $1^3 + 10^0$

a._____ **(2)**

b._____ **(2)**

3. **Below is a diagram of a rectangle ABCD.**



(a) Name the straight line BC.

(b). Which line is perpendicular to AB?

(c) Calculate the size of the angle marked Y.

a _____ **(1)**

b_____ **(1)**

c._____ **(2)**

4. A clock is showing 25 minutes after midday.

(a) Write this time in 12 hour clock.

(b) What time will it show after 2 hours 45 minutes?

_____ **(1)**

_____ **(2)**

5. **(a)** Add 0,765 ; 9 and 5,65.

(b) Sam wrote a test out of 40 and got 24 marks. Express this mark as a percentage.

_____ **(2)**

_____ **(2)**

6. **(a)** Simplify 0,618 ÷ 0,03.

(b) 100 as a product of prime factors.

_____ **(2)**

_____ **(2)**

7. The area of a triangle is $24cm^2$. Its base is 6cm. Calculate its height.

_____ **(3)**

8. Express 3 metres 20mm in millimetres.

_____ **(2)**

SECTION B (15 MARKS)

*Answer **any three** questions in this section*

9. Three boys shared $240. The first two shared equally 70% of the money. The rest was given to the the third boy.

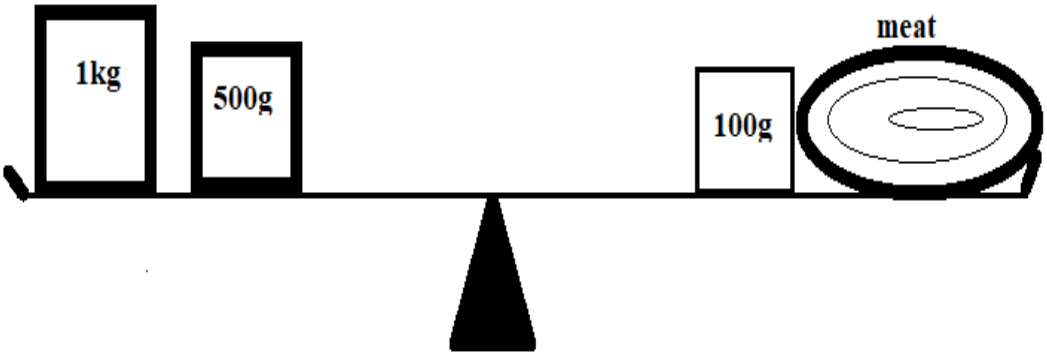
(a) What proper fraction of the total amount shared did the first boy receive?

(b) Find the share of the third boy.

a. _____ **(3)**

b._____ **(2)**

10. The diagram below shows meat on a scale.



(a) Write 100g in kilograms

(b) Find the mass of meat.

_____ **(2)**

_____ **(3)**

11. At a wedding, $\frac{7}{10}$ of a 50 kg bag of flour was used to make some buns. Each bun had a mass 70g.

a. **i.** Write $\frac{7}{10}$ as a decimal.

ii. How many buns were baked?

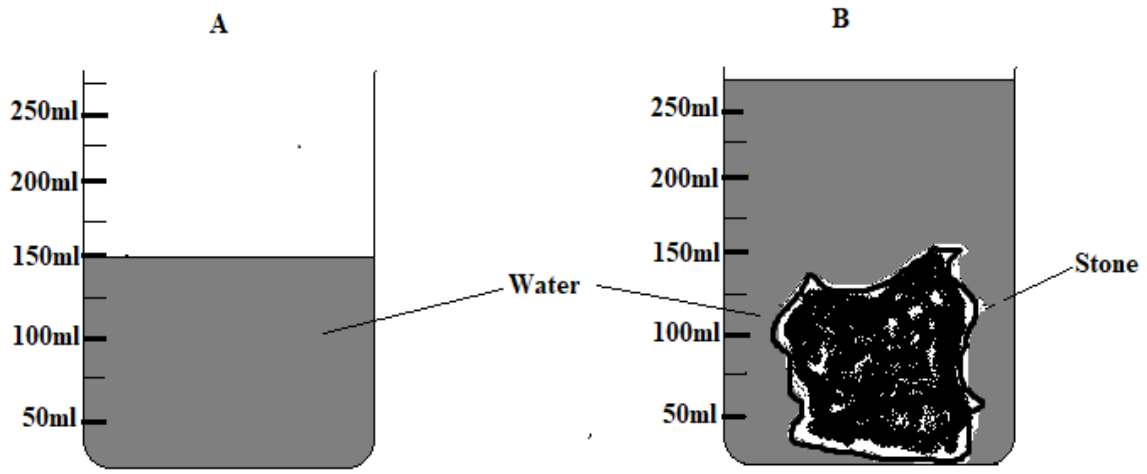
b. The baker used flour, butter and sugar in the ratio 4 : 2 : 1 respectively. 10kg of butter were used. Find the amount of flour used.

a. **i.**_____ **(1)**

ii_____ **(2)**

b. _____ **(2)**

12. A grade 7 boy used the following method to find volume of stone.



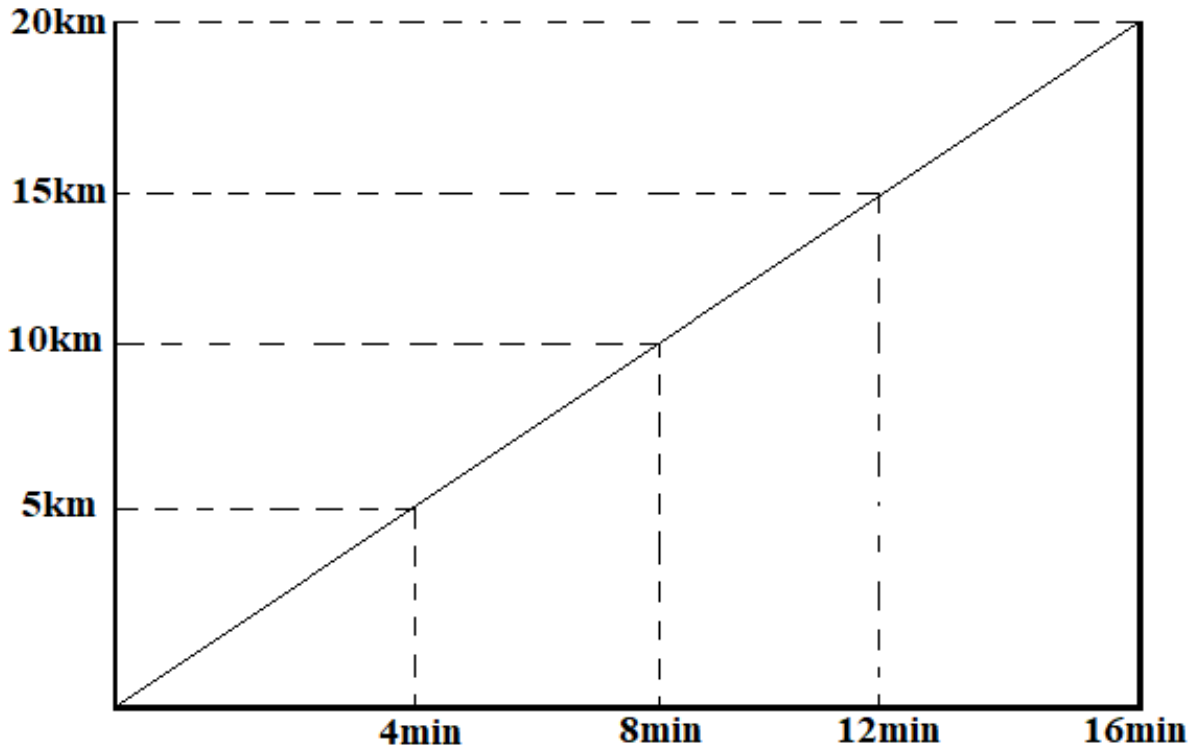
a. Write the amount of water in container A as a fraction of a litre in its lowest terms.

b. Giving your answer in cm^3, find the volume of the stone.

a._____ **(2)**

b._____ **(3)**

13. The graph below shows how a car travelled **20km.**



a. Name this type of graph.
b. How kilometres did this car cover in 8 minutes.
c. What was its average speed in kilometres per hour?

a. _____ **(1)**

b. _____ **(1)**

c. _____ **(3)**